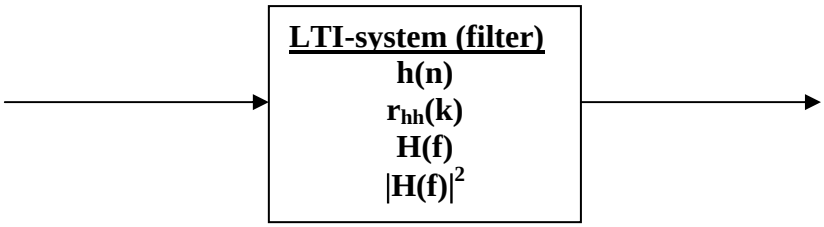
MATLAB-exercise: Stochastic signals and linear systems (filtering of stochastice signals)

0. The aim and course of the exercise

The purpose of the exercise is achieving knowledge of how a linear system (a filter) affects a stochastic signal, i.e. how correlatoin functions and power spectra are affected by the characteristics of the system. This is the foundation of several applications of stochastic signal processing, which are to be examined in coming exercises and can furthermore contribute to the physic/intuitive understanding of the correlation functions and power spectrum.

1. Collection of formulas for stochastic and linear systems



$x(n)$	(stoch. signal)	$y(n) = h(n) * x(n)$	(1a)
$m_x = E[x(n)]$	(aveage value)	$m_y = m_x \cdot H(0)$	(2a)
$\sigma_x^2 = E[(x(n) - m_x)^2]$	(effect)	$\sigma_y^2 = r_{yy}(0)$	(3a)
$r_{xx}(k) = E[x(n)x(n+k)]$	(autocorr.fct.)	$r_{yy}(k) = r_{hh}(k) * r_{xx}(k)$	(4a)
$P_{xx}(f) = F\{r_{xx}(k)\}$	(power spectrum)	$P_{yy}(f) = \lvert H(f)\rvert^2 P_{xx}(f)$	(5a)
	(crosscorr..fct.)	$r_{yx}(k) = h(k) * r_{xx}(k)$	(6a)
	(cross spectrum)	$P_{yx}(f) = H(f) P_{xx}(f)$	(7a)
		$r_{xy}(k) = h(-k) * r_{xx}(k)$	(8a)
		$P_{xy}(f) = H^*(f) P_{xx}(f)$	(9a)

Is particularly x(n) (aproximated) gauss-distributed white noise, then:

$x(n) = N(0, \sigma^2)$	(stoch. signal)	$y(n) = h(n) * x(n)$	(1b)
$m_x = 0$	(mean value)	$m_y = 0$	(2b)
$\sigma_x^2 = \sigma^2$	(effect)	$\sigma_y^2 = r_{yy}(0)$	(3b)
$r_{xx}(k) = \sigma^2 \cdot \delta(k)$	(autocorr.fct.)	$r_{yy}(k) = \sigma^2 \cdot r_{hh}(k)$	(4b)
$P_{xx}(f) = \sigma^2$	(power spectrum)	$P_{yy}(f) = \sigma^2 \cdot \lvert H(f)\rvert^2$	(5b)
	(cross corr. fct.)	$r_{yx}(k) = \sigma^2 \cdot h(k)$	(6b)
	(cross spectrum)	$P_{yx}(f) = \sigma^2 \cdot H(f)$	(7b)
		$r_{xy}(k) = \sigma^2 \cdot h(-k)$	(8b)
		$P_{xy}(f) = \sigma^2 \cdot H^*(f)$	(9b)

2. Input-signal

Gauss-distributed white noise is defines as
fs = 8000;
N = 1024;
n = 0:N-1;
sigma = 2;
x = sigma*randn(1,N);

Check the mean value and power with what is anticipated according to the collection of formulas mentioned above:
help mean
mean(x)
help std
std(x)^2

Check the autocorrelation function and power spectrum – both its appearance and scaling:
rxxPxxplot(x,'biased',fs)
Does this fit with the anticipated effect 10*log(sigma^2) = 6 dB ?
Save the figure for later usage.

3. System 1 (FIR MA LPF)

As system, first a simple Moving Average FIR LPF is used (also called "comb-filter")
h = 0.2*[1 1 1 1 1]; (average of the last 5 input-samples = simple LPF)
help impz
impz(h,1)
freqz(h,1,1024,fs)
zplane(h,1)
Compare the place of origin with the frequency characteristics.

The autocorrelation function for filter coefficients
[rhh kh] = xcorr(h,'none');
stem(kh,rhh,'.')

The stochastic input-signal is filtrated in the FIR-system:
y = filter(h,1,x);

Check the mean value for output-signal
mean(y)

Check the autocorrelation function, power and PSD for output signal, confer with the collection of formulas
figure(2)
rxxPxxplot(y,'biased',fs) use zoom-function for autocorr.
std(y)^2

Check the cross correlation function and cross spectrum between input and output, confer with the collection of formulas
rxyPxyplot(y,x,'biased',fs) use zoom-function
rxyPxyplot(x,y,'biased',fs)

4. System 2 (IIR BPF)

Now an IRR Butterworth BPF is used as system

help butter
[b a] = butter(4, [800/(fs/2) 1200/(fs/2)]
zplane(b,a)

Plot H(f), h(n) and $r_{hh}(k)$ for later use (park the figures)
freqz(b,a,1024,fs)
[h m] = impz(b,a);
stem(m,h,'.')
[rhh kh] = xcorr(h,'none');
stem(kh,rhh,'.') use zoom-function

The stochastic input-signal is filtrated in the FIR-system:
y = filter(b,a,x);

Check the mean value for output-signal
mean(y)

Check the autocorrelation function, power and PSD for output-signal, confer with the collection of formulas
rxxPxxplot(y,'biased',fs) use zoom-function for autocorr.
std(y)^2

Check the cross correlation function and cross spectrum between input and output, confer with the collection of formulas
rxyPxyplot(y,x,'biased',fs) use zoom-function
rxyPxyplot(x,y,'biased',fs)

5. Possibilities for further experiments

If time allows it, it is possible to experiment with other systems and/or other types of input-signals, so that the formulas (1a)-(9a) mentioned above, will be activated instead of (1b)-(9b).